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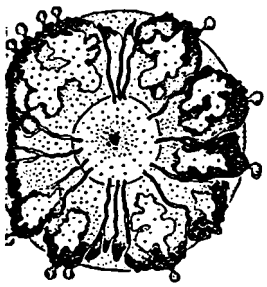
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Palaeomagnetism of dykes along the southern margin of the Baltic Shield

GÖRAN BYLUND



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217 cores from 14 quartz dolerite and 12 melaphyre dykes found at the southern margin of the Baltic Shield in southern Sweden gave mean palaeomagnetic pole positions at 38.5°N., 168.5°E. and 54°N., 171.5°E. respectively. The quartz dolerite pole position agrees with the Upper Carboniferous palaeomagnetic pole for W. Europe, while the position of the melaphyre pole is in better agreement with the Permian palaeomagnetic pole. The result indicates the existence of two different, late Palaeozoic episodes of hypabyssal igneous activity along the southern boundary of the Baltic Shield.

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In Skåne, southernmost Sweden, dykes follow tectonic lineaments which run in a northwest–southeasterly direction. These tectonic lines, which are mostly faults, mark the southern delimitation of the Precambrian Baltic Shield. The natural remanent magnetization (NRM) of six quartz dolerite dykes in this part of the marginal zone of the shield has previously been studied by Mulder (1971). His results indicate an Upper Carboniferous dyke age. Apart from the quartz dolerites, there exists in central Skåne another set of more basic dykes, the melaphyres, which have the same northwesterly direction of strike. The dissimilarity of composition may suggest different periods of igneous and tectonic activity along the southern margin of the Baltic Shield. In an attempt to solve this problem by continuing the study of the NRM in the Skåne dykes, orientated samples were collected from 14 quartz dolerite and 12 melaphyre dykes.

Precambrian gneisses and granite gneisses prevail in the northern half of Skåne, the rest being covered by younger sediments (map Fig. 1).

The Post-Silurian tectonics are dominated by northwesterly faults, and subordinately a less well developed set of north–northeast trending dislocation lines. No tectonic rotation of any importance for this study has been discovered.

The quartz dolerites are composed of labradorite, augite, ore, apatite, and micrographic intergrowths of quartz and alkali feldspar. Some dykes are olivine-bearing. The melaphyres are porphyrites, carrying phenocrysts of olivine and augite and exhibiting amygdaloidal textures. The matrix consists of varying proportions of plagioclase, pyroxene, hornblende and glass. The phenocrysts may be replaced by serpentine and chlorite. Generally, the melaphyres are severely altered, so that it is often difficult to discern the original constituents. This alteration has been ascribed to postintrusion autometamorphism due to high content of volatiles in the original melaphyre magma (Hjelmqvist 1939). The quartz dolerites are generally well preserved.

There is no general agreement on the age of the dyke intrusions. Hjelmqvist (1939) holds that the whole dyke system is Lower Permian and connected with the igneous activity in the Oslo area. Mulder's palaeomagnetic measurements, mentioned above, indicate an Upper Carboniferous quartz dolerite age. Due to petrological similarities, Törnebohm & Hennig (1904), and later Printzlau & Larsen (1972), suggested a connection between a group of basalts in central Skåne and the melaphyres. The basalts are considered to be Jurassic to Cretaceous (Printzlau & Larsen 1972, Klingspor 1973).

Opaque petrography

The opaques in the quartz dolerites and the melaphyres were studied by reflected light microscopy and X-ray diffraction. The reflected light study demonstrated the presence in the quartz dolerites of titanomagnetite with high-oxidation lamellae and patches of ilmenite. No signs of

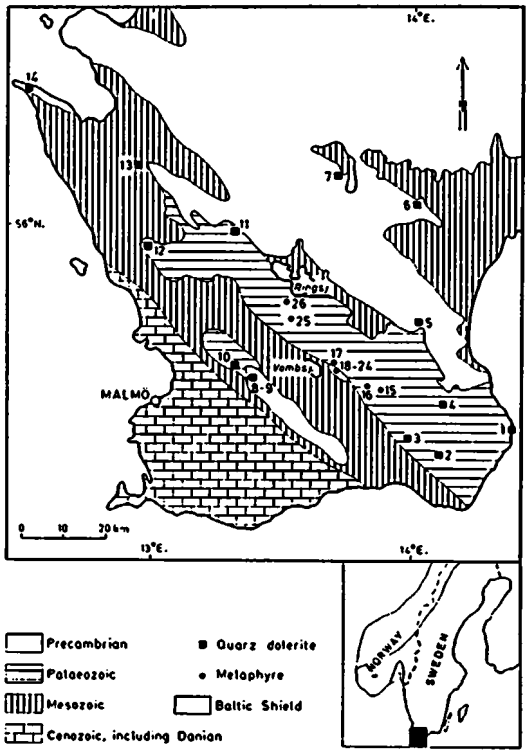


Fig. 1. Geologic map of Skåne with sampling sites.

high-temperature oxidation were found in the melaphyre titanomagnetite. The melaphyres carry secondary magnetite formed by the alteration of the primary ferromagnesian minerals. This type of magnetite is found in clusters of fine opaque grains, forming rims and crack-fillings in pseudomorphs after olivine and pyroxene. Insignificant amounts of hematite were observed. One melaphyre dyke, number 17, contains a large amount of hematite and limonite but almost no magnetite. The hematite here partly replaces original magnetite and partly occurs together with irregular limonite masses. The formation of hematite in this dyke is probably a two-stage process, implying oxidation of original magnetite as well as a later formation of hematite due to superficial effects.

The X-ray diffraction curves obtained are complex, but both rock types show peaks due to a cubic phase having lattice parameters between 8.39 and 8.50 Å. This suggests members of the magnetite-ulvospinel series (Basta 1960). Additional peaks found in the melaphyre curves indicate the presence of hematite and titanium oxides. The X-ray study thus roughly confirms the optical results.

Sampling, measurement and results

Orientated core samples were obtained by drilling with a small portable core drill. Each dyke provided a minimum of six cores. The available exposures varied from one meter to one kilometer and more in length. Some dykes were therefore sampled only at one locality, whereas others permitted the collection of representative samples from up to five different sites. No significant differences were found between the results from different sites in the same dyke, and thus each dyke is treated as one single unit in the calculations. Seven of the melaphyre dykes were sampled in one quarry containing a dyke swarm (dykes numbers 18–24). Moreover most of the remaining sampling was done in road cuttings and quarries.

The NRM was measured by a four-probe fluxgate magnetometer (Förster Oerstedmeter) and a spinner magnetometer of Czechoslovakian make (UGF-JR3). The initial NRM in the quartz dolerites ranged from $1 \cdot 10^{-3}$ to $8 \cdot 10^{-3}$ emu/cm^3, and in the melaphyres from $1 \cdot 10^{-5}$ to $50 \cdot 10^{-5}$ emu/cm^3. The quartz dolerites had Q-values well above unity, while the Q-values of the melaphyres were generally below unity.

The alternating-field technique was used for removal of soft viscous magnetic components. From each drilling site at least one pilot specimen was treated in demagnetization steps of 50 Oe, continued to peak fields of 400–800 Oe. The remaining samples were then treated in the appropriate alternating fields. The behaviour during demagnetization was similar for both rock types. The NRM decreased to about 50 % of the original value after stepwise demagnetization to peak fields of 200–300 Oe. The amount of directional change varied, but consistent directions appeared after demagnetization to 200–300 Oe. Examples of the behaviour of pilot specimens during demagnetization are given in Fig. 2. The specimens from one quartz dolerite dyke (number 3) and three melaphyre dykes (numbers 15, 16, and 21) proved unstable. They displayed a large scatter of directions both within and between specimens during the stepwise demagnetization. They also showed low resistance against increasing demagnetizing fields as exemplified by curves 3 and 15 in Fig. 2*b* and 2*d*. Specimens from melaphyre dyke number 17 rendered high coercivity curves, exemplified by curve 17 in Fig. 2*d*. This implies that the secondarily formed hematite in this dyke is the main carrier of its NRM. The magnetic direction found in this dyke deviates significantly from the other melaphyre directions and is not included in the cal-

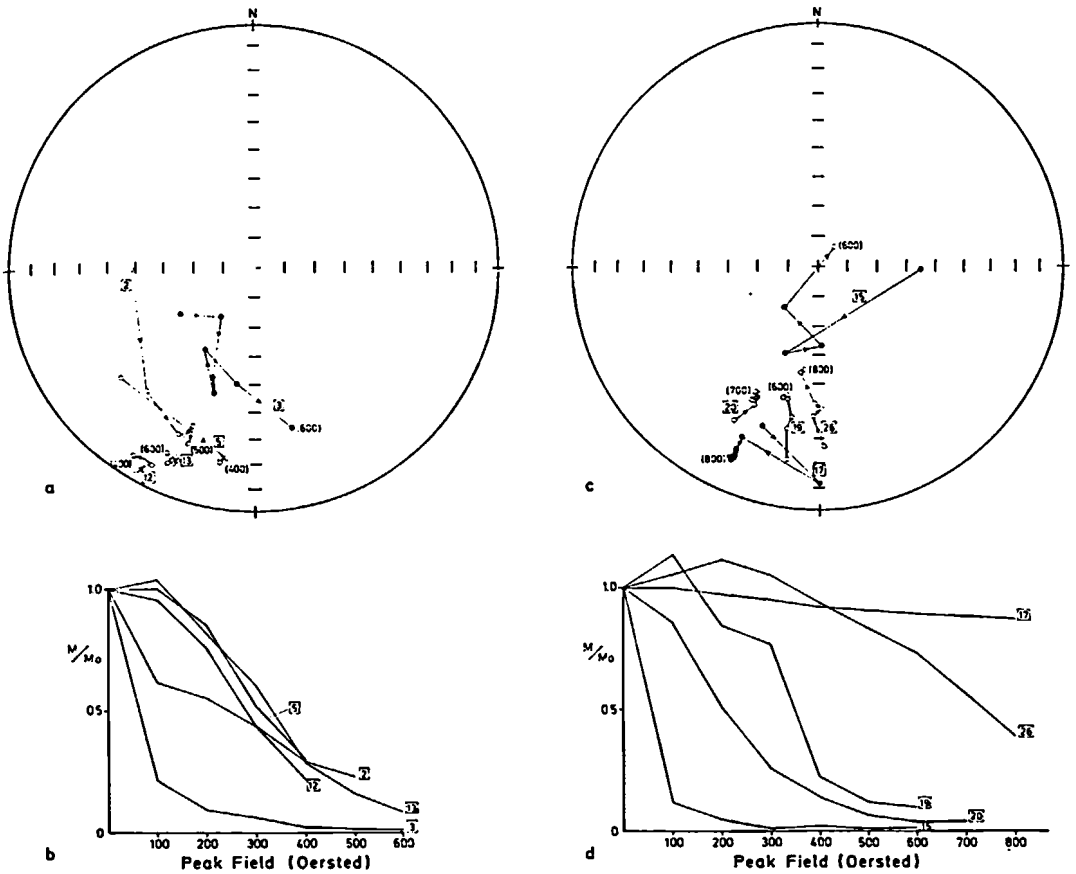


Fig. 2. Examples of the behaviour of quartz dolerites during AF-demagnetization are shown in *a* and *b*; *c* and *d* give the corresponding data for the melaphyres. Open circles: upward north-seeking magnetization, filled circles: downward north-seeking magnetization. Figures in squares refer to dyke number, those in brackets to peak alternating fields used for demagnetization.

culation of the melaphyre mean. Mean NRM directions for each demagnetization step were calculated according to Fisher (1953). Table 1 lists the mean directions from each dyke, presented on the demagnetization level with the least scatter as given by the statistical parameters k and α_{95}.

The pole positions obtained were 38.5°N., 168.5°E. for the quartz dolerites and 54°N., 171.5°E. for the melaphyres.

Discussion

There is a significant difference in magnetic directions between the quartz dolerites and the melaphyres (Fig. 3). The calculated quartz dolerite pole agrees well with the Upper Carboniferous poles from neighbouring areas in stable

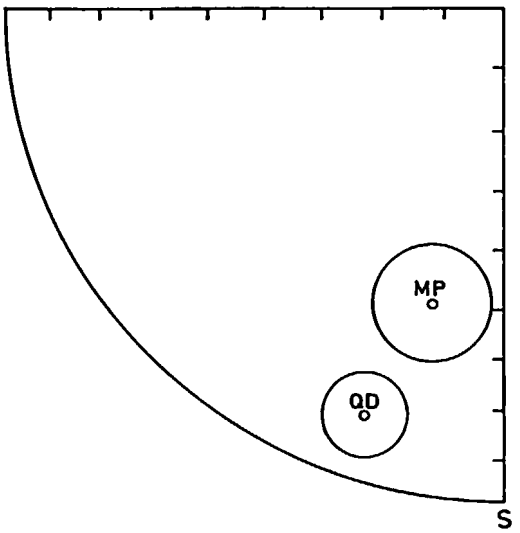


Fig. 3. Mean magnetic directions for quartz dolerites (QD) and melaphyres (MP) with 95 % circles of confidence. Dyke means were used for the calculation of rock-type mean directions.

Table 1. Palaeomagnetic results after AF-demagnetization. Dykes numbers 1–14 are quartz dolerites, numbers 15–26 are melaphyres.

Dyke	Location (°N)	Location (°E)	*a/b*	AF (Oe)	*D* (°)	*i* (°)	*k*	α_{95} (°)
1	55.5	14.4	4/6	300	190	— 1.5	53	12.5
2	55.5	14.0	6/6	200	205.5	—25	127	5.5
3	55.5	13.9	0/7	—	—	—	—	—
4	55.6	14.1	4/6	300	200	—17	472	4
5	55.8	14.2	6/6	300	191	—22.5	292	4
6	56.1	14.0	6/6	200	202.5	—29	90	7
7	56.1	13.7	6/6	200	215	— 8.5	21	15
8	55.6	13.4	17/18	200	202.5	—16.5	39	6
9	55.6	13.4	6/6	200	193	— 7	339	3.5
10	55.7	13.3	12/12	200	191	—22	195	3
11	56.0	13.3	12/12	200	194.5	— 3	68	5
12	56.0	13.0	15/15	300	202.5	—13	66	5
13	56.1	12.9	6/6	200	188	— 7	30	12.5
14	56.3	12.5	6/6	200	219	—21	122	6
Mean: 13 quartz dolerite dykes					199.5	—15	40	6.5
Mean: 106 quartz dolerite specimens					199.5	—15	30	2.5
15	55.7	13.8	0/5	—	—	—	—	—
16	55.7	13.8	0/4	—	—	—	—	—
17	55.7	13.7	29/30	400	212	+17.5	14	4.5
18	55.7	13.7	4/4	100	208	—14.4	906	3
19	55.7	13.7	6/6	100	199	—25.5	372	3.5
20	55.7	13.7	5/5	200	202.5	—37.5	451	3.5
21	55.7	13.7	0/5	—	—	—	—	—
22	55.7	13.7	11/11	100	191	—49	142	4
23	55.7	13.7	7/7	200	190	—32	1923	1.5
24	55.7	13.7	7/7	200	184.5	—59	377	3
25	55.8	13.5	4/6	300	180	—33	88	10
26	55.8	13.5	5/7	300	178.5	—47	79	9
Mean: 9 melaphyre dykes[1]					193.5	—38	26	11
Mean: 49 melaphyre specimens[1]					193	—41	29	4

a: number of specimens used in the calculation of the dyke mean direction, *b:* number of specimens measured, AF: peak demagnetizing field used, *D:* declination, *i:* inclination, *k:* precision parameter, α_{95}: radius of the 95 % circle of confidence.

[1]) Dyke 17 is not included in the calculation of the melaphyre mean.

Europe listed in Table 2. This circumstance and the consistent directions and stable behaviour during demagnetization imply that the stable NRM observed in the quartz dolerites is the original thermal remanent magnetization (TRM). The melaphyre pole is closer to the Permian and Triassic poles as listed in Table 2. There is a possibility that part of the stable NRM in the melaphyres is caused by a secondary chemical remanent magnetization (CRM), due to the formation of magnetic phases during the breakdown of original ferromagnesian minerals. If this process is a result of autometamorphism caused by high amounts of volatiles in the melaphyre magma as suggested by Hjelmqvist (1939), then the time difference between the formation of the original TRM and the secondary CRM is insignificant. The stable NRM found in the melaphyres then approximately represents the NRM acquired during the primary formation of the rocks. The conclusion is that the quartz dolerites and the melaphyres represent two different episodes of igneous activity along the border of the Baltic Shield. The quartz dolerite data indicate an Upper Carboniferous intrusion age, whereas the melaphyres are younger and probably Permian. However, the possibility of a Triassic age cannot be excluded on the basis of the available data.

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Table 2. Magnetic poles from stable Europe used for comparison with the results from the Scanian dykes.

Rock formation or area	Age	Pole position		Reference
Dolerite dykes, Skåne, Sweden	Cu	37°N.	174°E.	Mulder 1971
Dolerite sill, E. Västergötland, Sweden	Cu	31°N.	174°E.	Mulder 1971
Dolerite sill, W. Västergötland, Sweden	Cu	38°N.	166°E.	Mulder 1971
Ny-Hellesund diabase, Norway	Cu–Pl	39°N.	160°E.	Halvorsen 1970
Mean of W. Europe and Russian platform	Cu	37°N.	167°E.	McElhinny 1973
Quartz dolerites, Skåne, Sweden		38.5°N.	168.5°E.	This paper
Oslo igneous rocks, Norway	Pl	47°N.	157°E.	van Everdingen 1960
Krakow volcanics, Poland	Pl	43°N.	164°E.	Birkenmajer & Nairn 1964
Volcanics of inner Sudetic basin, Poland	Pl	44°N.	167°E.	Birkenmajer *et al.* 1968
Igneous rocks, Black Forest, Germany	P	48.5°N.	174°E.	Konrad & Nairn 1972
Mean of W. Europe and Russian platform	P	44°N.	161°E.	McElhinny 1973
Mean of W. Europe and Russian platform	Tr	50°N.	150°E.	McElhinny 1973
Melaphyres, Skåne, Sweden		54°N.	171.5°E.	This paper

Code: C: Carboniferous, P: Permian, Tr: Triassic, u: upper, l: lower.

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